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Collected Mathematical Papers, Vol. I
New work: book pp. 441–457

ON A THEORY OF THE SYZYGETIC RELATIONS OF TWO RATIONAL
INTEGRAL FUNCTIONS, COMPRISING AN APPLICATION TO THE THEORY
OF STURM'S FUNCTIONS, AND THAT OF THE GREATEST ALGEBRAICAL
COMMON MEASURE

[*Philosophical Transactions of the Royal Society, CXLIII. (1853), pp. 407-548*]

p. 441

but $N = 0$ makes ω vanish, and therefore, upon this supposition, f and ϕ would appear to have a common algebraical factor ψ , that is to say, N vanishing would appear to imply that the resultant of f and ϕ must vanish, so that N would appear to be contained as a factor in this general resultant, which latter is, however, clearly indecomposable into factors—a seeming paradox—the solution of which must be sought for in the fact, that the equation $N = 0$ is *incompatible* with the existence of the usual equations (7) connecting f, ϕ, ψ and ω ; but this failure of the existence of the equations (7) (bearing in mind that N has been shown to be a function only of the set of coefficients $a, \beta, \&c.$), can only happen by reason of a vanishing whenever N vanishes; a must therefore be a root of N , or which is the same thing, N a power of a and hence $N = a^2$.

The same result may be obtained *à posteriori* by actually performing the successive divisions; if the coefficients of any dividend be $a, b, c, d, \&c.$, and of the divisor $\alpha, \beta, \gamma, \delta, \&c.$, the first remainder, forming the second divisor, will be easily seen to have for its coefficients—

$$\frac{1}{a^2} \begin{vmatrix} a & b & c \\ 0 & \alpha & \beta \\ \alpha & \beta & \gamma \end{vmatrix}, \quad \frac{1}{a^2} \begin{vmatrix} a & b & d \\ 0 & \alpha & \beta \\ \alpha & \beta & \delta \end{vmatrix}, \quad \frac{1}{a^2} \begin{vmatrix} a & b & e \\ 0 & \alpha & \beta \\ \alpha & \beta & \varepsilon \end{vmatrix} \quad \&c.$$

Hence the coefficients in the next remainder (making $\begin{vmatrix} a & b & c \\ 0 & \alpha & \beta \\ \alpha & \beta & \gamma \end{vmatrix} = m$) will be each of the form of the compound determinant,—

$$\frac{1}{m^2} \begin{vmatrix} \alpha & \beta & \gamma \\ 0 & \begin{vmatrix} a & b & c \\ 0 & \alpha & \beta \\ \alpha & \beta & \gamma \end{vmatrix} & \begin{vmatrix} a & b & d \\ 0 & \alpha & \gamma \\ \alpha & \beta & \delta \end{vmatrix} \\ \begin{vmatrix} a & b & c \\ 0 & \alpha & \beta \\ \alpha & \beta & \gamma \end{vmatrix} & \begin{vmatrix} a & b & d \\ 0 & \alpha & \gamma \\ \alpha & \beta & \delta \end{vmatrix} & \begin{vmatrix} a & b & e \\ 0 & \alpha & \beta \\ \alpha & \beta & \varepsilon \end{vmatrix} \end{vmatrix}.$$

The compound determinant above written will be the first coefficient in the remainder under consideration; the subsequent coefficients will be represented by writing $f, \phi; g, \gamma, \&c.$ respectively in lieu of e, ε . Omitting the common multiplier $\frac{1}{m^2}$, the determinant above written is equal to

p. 442

$$\alpha \left\{ \begin{vmatrix} a & b & c \\ 0 & \alpha & \beta \\ \alpha & \beta & \gamma \end{vmatrix} \begin{vmatrix} a & b & e \\ 0 & \alpha & \delta \\ \alpha & \beta & \varepsilon \end{vmatrix} - \begin{vmatrix} a & b & d \\ 0 & \alpha & \gamma \\ \alpha & \beta & \delta \end{vmatrix} \begin{vmatrix} a & b & d \\ 0 & \alpha & \gamma \\ \alpha & \beta & \delta \end{vmatrix} \right\} \\ + \begin{vmatrix} a & b & c \\ 0 & \alpha & \beta \\ \alpha & \beta & \gamma \end{vmatrix} \left\{ \beta \begin{vmatrix} a & b & d \\ 0 & \alpha & \gamma \\ \alpha & \beta & \delta \end{vmatrix} - \gamma \begin{vmatrix} a & b & c \\ 0 & \alpha & \beta \\ \alpha & \beta & \gamma \end{vmatrix} \right\}.$$

The last written pair of terms are together equal to

$$\begin{vmatrix} a & b & c \\ 0 & \alpha & \beta \\ \alpha & \beta & \gamma \end{vmatrix} \times \{-d\beta a^2 + c\gamma a^2 + a\alpha(\beta\delta - \gamma^2)\},$$

which is of the form $a^2A - a^2\beta^2(\beta\delta - \gamma^2)\alpha$; and the sum of the first written pair is of the form $a^2B + (a\beta^2 a\beta\delta - a\gamma\beta a\gamma\beta)\alpha$. Hence the entire determinant is of the form $a^2(A + B)$, showing that a^2 will enter as a factor into this and every subsequent coefficient in the second remainder, as previously demonstrated above.

It may, moreover, be noticed, that this remainder, when a^2 has been expelled, will for general values of the coefficients be numerically as well as literally in its lowest terms, as evinced by the fact that there exist terms (for example $a\alpha^2\gamma\epsilon$) having +1 for their numerical part. The same explicit method might be applied to show, that if the first divisor were e degrees instead of being only one degree in x lower than the first dividend, a^{e+1} would be contained in every term of the second residue: the difficulty, however, of the proof by this method augments with the value of e ; but the same result springs as an immediate consequence from the method first given, which remains good *mutatis mutandis* for the general case, as may easily be verified by the reader. Applying now this result to the functions P and Q , supposed to be of the respective degrees n and $n - e$ in x , and calling the coefficients of the leading terms in the successive simplified residues $a_1, a_2, a_3, \&c.$, and denoting by a the leading coefficient in Q , and as before denoting the successive allotrious factors by $\lambda_1, \lambda_2, \&c.$, it will readily be seen that

$$\lambda_1 = \frac{1}{a^{e+1}}, \quad \lambda_2\lambda_1 = \frac{1}{a_1^2}, \quad \lambda_3\lambda_2 = \frac{1}{a_2^2}, \quad \lambda_4\lambda_3 = \frac{1}{a_3^2}, \quad \&c.,$$

that is

$$\lambda_1 = \frac{1}{a^{e+1}}, \quad \lambda_2 = \frac{a^{e+1}}{a_1^2}, \quad \lambda_3 = \frac{a_1^2}{a^{e+1}a_2^2}, \quad \lambda_4 = \frac{a^{e+1}a_2^2}{a_1^2a_3^2},$$

and in general

$$\left. \begin{aligned} \lambda_{2m+1} &= \frac{1}{a^{e+1}} \frac{a_1^2 a_3^2 \cdots a_{2m-1}^2}{a_2^2 a_4^2 \cdots a_{2m}^2}, \\ \lambda_{2m} &= a^{e+1} \frac{a_2^2 a_4^2 \cdots a_{2m-2}^2}{a_1^2 a_3^2 \cdots a_{2m-1}^2}. \end{aligned} \right\} \quad (8)$$

p. 443

Art. (4). Strictly speaking, we have not yet fully demonstrated that the complete allotrious factors are represented by the values above given for λ , but only that these latter are contained as factors in the allotrious factors; we must further prove that there exist no other such factors. This may be shown as follows: it is obvious from the nature of the process that the complete residues will always

remain of one dimension in respect of the given coefficients, that is, first of one dimension in the set $a, b, c, \&c.$, and of zero dimensions in $\alpha, \beta, \gamma, \&c.$; then conversely, of one dimension in $\alpha, \beta, \gamma, \&c.$, and of zero dimensions in $a, b, c, \&c.$, and so on, the residues being evidently required to conform in their dimensions to those of the first dividend and the first divisor alternately. These coefficients then are always of unit dimensions in respect to the given coefficients; whereas it has been shown (Art. 2) that the simplified residues in respect to these coefficients are successively of the dimensions $2 + e, 4 + e, 6 + e, \&c.$

Let the complete residue corresponding to λ_{2m} be $M\lambda_{2m}a_{2m}$, that is

$$M \frac{a^{e+1}}{a_1^2} \frac{a_2^2}{a_3^2} \frac{a_4^2}{a_5^2} \cdots \frac{a_{2m-2}^2}{a_{2m-1}^2} a_{2m},$$

or say ML ; in passing from a_{2q} to a_{2q+1} the dimensions rise 2 units for all values of q except zero, and when $q = 0$ the dimensions increase *per saltum* from 1 to $2 + e$; hence the total dimensions of L in the joint coefficients will be

$$\{(e + 1) - 2(e + 2)\} - 4(m - 1) + 4m + e = 1,$$

and therefore M is of zero dimensions, and λ_{2m} is the complete allotrious factor. In like manner if the complete residue corresponding to λ_{2m+1} be $M\lambda_{2m+1}a_{2m+1}$, that is

$$M \frac{1}{a^{e+1}} \frac{a_1^2}{a_2^2} \frac{a_3^2}{a_4^2} \cdots \frac{a_{2m-1}^2}{a_{2m}^2} a_{2m+1},$$

or say ML , the dimensions of L will be

$$-(e + 1) - 4m + \{e + 2(2m + 1)\}, \quad \text{that is, } 1,$$

and hence, as in the preceding case, M is of zero dimensions, and λ_{2m+1} is the complete allotrious factor.

Art. (5). I proceed to show how the simplified residues may be most conveniently obtained by a direct process, identical with that which comes into operation in applying to the two given functions of x the method familiarly known under the name of Bezout's abridged method of elimination. Let us call the two given functions U and V , and commence with the case where U and V are of equal dimensions (n) in x . The simplified ι th residue will then be a function of $n - \iota$ dimensions in x , and of ι dimensions in respect of each given set of coefficients, and may be taken equal to $V_\iota U + U_\iota V$, where V_ι and U_ι are each of $(\iota - 1)$ dimensions in x .

p. 444

Let

$$U = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n,$$

$$V = b_0 x^n + b_1 x^{n-1} + b_2 x^{n-2} + \cdots + b_n;$$

we may write in general, m being taken any positive integer not exceeding n ,

$$U = (a_0x^m + a_1x^{m-1} + \cdots + a_m)x^{n-m} + (a_{m+1}x^{n-m-1} + a_{m+2}x^{n-m-2} + \cdots + a_n),$$

$$V = (b_0x^m + b_1x^{m-1} + \cdots + b_m)x^{n-m} + (b_{m+1}x^{n-m-1} + b_{m+2}x^{n-m-2} + \cdots + b_n).$$

Hence

$$(b_0x^m + b_1x^{m-1} + \cdots + b_m)U - (a_0x^m + a_1x^{m-1} + \cdots + a_m)V$$

$$= {}_mK_1x^{2n-1} + {}_mK_2x^{2n-2} + {}_mK_3x^{2n-3} + \cdots + {}_mK_n, \quad (9)$$

where if we use (r, s) to denote $a_rb_s - a_sb_r$ for all values of r and s , we have

$${}_mK_1 = (0, m+1), \quad {}_mK_2 = (0, m+2) + (1, m+1),$$

$${}_mK_3 = (0, m+3) + (1, m+2) + (2, m+1),$$

and in general ${}_mK_i = \Sigma(r, s)$, the values of r and s admissible within the sign of summation being subject to the two conditions, one the equality $r + s = m + i$, the other the inequality r less than i . By giving to m all the different values from 0 to $m - 1$ in succession, and calling

$$b_0x^m + b_1x^{m-1} + \cdots + b_m, \quad a_0x^m + a_1x^{m-1} + \cdots + a_m$$

respectively Q_m and P_m , we have

$$\left. \begin{aligned} Q_0U - P_0V &= K_1x^{2n-1} + K_2x^{2n-2} + \cdots + K_n, \\ Q_1U - P_1V &= {}_1K_1x^{2n-1} + {}_1K_2x^{2n-2} + \cdots + {}_1K_n, \\ Q_2U - P_2V &= {}_2K_1x^{2n-1} + {}_2K_2x^{2n-2} + \cdots + {}_2K_n, \\ &\dots\dots\dots \\ Q_{n-1}U - P_{n-1}V &= {}_{n-1}K_1x^{2n-1} + {}_{n-1}K_2x^{2n-2} + \cdots + {}_{n-1}K_n. \end{aligned} \right\} \quad (10)$$

The right-hand members of these n equations I shall henceforth term the Bezoutians to U and V .

The determinant formed by arranging in a square the n sets of coefficients of the n Bezoutians, and which I shall term the Bezoutian matrix, gives, as is well known, the Resultant (meaning thereby the Result in its simplest form of eliminating the variables out) of U and V .

Eliminating dialytically, first x^{n-1} between the first and second, then x^{n-1} and x^{n-2} between the first, second and third, and so on, and finally, all the powers of x between the first, second, third, $\dots$ n th of these Bezoutians, and repeating the first of them, we obtain a derived set of n equations, the right-hand members of which I shall term the secondary Bezoutians to U and V , this secondary system of equations being

$$\begin{aligned}
Q_0U - P_0V &= K_1x^{2n-1} + K_2x^{2n-2} + K_3x^{2n-3} + \cdots + K_n, \\
({}_1K_1Q_0 - K_1Q_1)U - ({}_1K_1P_0 - K_1P_1)V \\
&= L_1x^{2n-2} + L_2x^{2n-3} + \cdots + L_{n-1}, \\
\{({}_1K_{12}K_2 - {}_2K_{11}K_2)Q_0 + ({}_2K_1K_2 - K_{12}K_2)Q_1 \\
&\quad + (K_{11}K_2 - {}_1K_1K_2)Q_2\}U \\
&\quad - \{({}_1K_{12}K_2 - {}_2K_{11}K_2)P_0 + ({}_2K_1K_2 - K_{12}K_2)P_1 \\
&\quad + (K_{11}K_2 - {}_1K_1K_2)P_2\}V \\
&= M_1x^{2n-3} + M_2x^{2n-4} + \cdots + M_{n-2}, \\
&\&c. = \&c.
\end{aligned} \tag{11}$$

And we can now already without difficulty establish the important proposition, that the successive simplified residues to $\frac{U}{V}$, expanded under the form of an improper continued fraction, abstracting from the algebraical sign (the correctness of which also will be established subsequently), will be represented by the n successive *Secondary Bezoutians* to the system U, V .

For if we write the system of equations (11) under the general form

$$\mathfrak{D}_\iota U - H_\iota V = A_\iota x^{n-\iota} + B_\iota x^{n-\iota-1} + \&c.,$$

the degree of $\mathfrak{D}_\iota$ and H_ι in x will be that of $Q_{\iota-1}$ and $P_{\iota-1}$, that is $\iota - 1$; and the dimensions of $A_\iota, B_\iota, \&c.$, in respect of each set of coefficients is evidently ι ; consequently, by virtue of Art. 2, $A_\iota x^{n-\iota} + B_\iota x^{n-\iota-1} + \&c.$, which is the ι th Bezoutian, will (saving at least a numerical factor of a magnitude and algebraical sign to be determined, but which, when proper conventions are made, will be subsequently proved to be +1) represent the ι th simplified residue to $\frac{U}{V}$,¹ as was to be shown.

Art. (6). More generally, suppose U and V to be respectively of $n + e$ and n dimensions in x .

Let

$$U = a_0x^{n+e} + a_1x^{n+e-1} + a_2x^{n+e-2} + \&c., \quad V = b_0x^n + b_1x^{n-1} + \&c.$$

Making

$$\begin{aligned}
U &= (a_0x^{e+m} + a_1x^{e+m-1} + \&c. + a_{e+m})x^{n-m} \\
&\quad + (a_{e+m+1}x^{n-m-1} + \&c. + a_{n+e}), \\
V &= (b_0x^m + b_1x^{m-1} + \cdots + b_m)x^{n-m} + (b_{m+1}x^{n-m-1} + \&c. + b_n),
\end{aligned}$$

¹ V is supposed to be taken as the first divisor, and the term residue is used, as hitherto in this paper, throughout in the sense appertaining to the expansion conducted, so as to lead to an *improper* continued fraction, in that sense, in fact, in which it would, more strictly speaking, be entitled to the appellation of *excess* rather than that of residue.

we obtain the equation

$$Q_m U - P_{e+m} V = {}_m K_1 x^{n+e-1} + {}_m K_2 x^{n+e-2} + \&c. + {}_m K_{n+e}, \quad (12)$$

p. 446

where

$$\begin{aligned} Q_m &= (b_0 x^m + \cdots + b_m), & P_{e+m} &= (a_0 x^{e+m} + \cdots + a_{e+m}); \\ {}_m K_1 &= a_0 b_{m+1}, \\ {}_m K_2 &= a_0 b_{m+2} + a_1 b_{m+1}, \quad \cdots, \\ {}_m K_e &= a_0 b_{m+e} + a_1 b_{m+e-1} + \&c. + a_e b_m, \\ {}_m K_{e+1} &= a_0 b_{m+e+1} + \&c. + a_{e+1} b_m - a_{e+m+1} b_0, \quad \&c. = \&c. \end{aligned}$$

By giving to m every integer value from 0 to $(n-1)$ inclusive, we thus obtain n equations of the form of (12), each of the degree $n+e-1$ in x , and of one dimension in regard to each set of coefficients.

In addition to these equations we have the e equations of the form

$$x^\mu V = b_0 x^{n+\mu} + b_1 x^{n+\mu-1} + \&c. + b_n x^\mu, \quad (13)$$

in which μ may be made to assume every value from 0 to $(e-1)$ inclusive, and the right-hand side of the equation for all such values of μ will remain of a degree in x not exceeding $n+e-1$, the degree of the equations of the system above described. There will thus be e equations in which only the (b) set of coefficients appear, and n equations containing in every term one coefficient out of each of the two sets.

The total number of equations is of course $n+e$. Between the e equations of the second system (13) and the r occurring first in order of the first system (12), we may eliminate dialytically the $e+r-1$ highest powers of x , and there will thus arise an equation of the form

$$\theta_{r-1} U - \omega_{e+r-1} V = L x^{n-r} + L' x^{n-r-1} + \&c. + (L), \quad (14)$$

where θ_{r-1} and ω_{e+r-1} are respectively of the degrees $r-1$ and $e+r-1$ in x , and $L, L' \dots (L)$ are of r dimensions in the (a) set, and of $(e+r)$ dimensions in the (b) set of coefficients, and consequently $L x^{n-r} + L' x^{n-r-1} + \cdots + (L)$ must satisfy the conditions necessary and sufficient to prove its being (to a numerical factor *près*) a simplified residue to (U, V) .

Thus suppose

$$U = a_0 x^4 + a_1 x^3 + a_2 x^2 + a_3 x + a_4, \quad V = b_0 x^2 + b_1 x + b_2.$$

Then, corresponding to the system of which equation (13) is the type, we have

$$V = b_0 x^2 + b_1 x + b_2, \quad xV = b_0 x^3 + b_1 x^2 + b_2 x.$$

Again, to form the system of which equation (12) is the type, we write

$$\begin{aligned}
b_0U - (a_0x^2 + a_1x + a_2)V &= b_0(a_3x + a_4) - (a_0x^2 + a_1x + a_2)(b_1x + b_2) \\
&= -a_0b_1x^3 - (a_0b_2 + a_1b_1)x^2 \\
&\quad + (b_0a_3 - a_1b_2 - a_2b_1)x + (b_0a_4 - a_2b_2), \\
(b_0x + b_1)U - (a_0x^3 + a_1x^2 + a_2x + a_3)V &= (b_0x + b_1)a_4 - (a_0x^3 + a_1x^2 + a_2x + a_3)b_2 \\
&= -a_0b_2x^3 - a_1b_2x^2 \\
&\quad + (b_0a_4 - a_2b_2)x + (b_1a_4 - b_2a_3).
\end{aligned}$$

Combining the two equations of the first system with the first of the second system, we obtain the first simplified residue $Lx + L'$, where p. 447

$$-L = \begin{vmatrix} 0 & b_0 & b_1 \\ b_0 & b_1 & b_2 \\ a_0b_1 & a_0b_2 + a_1b_1 & a_1b_2 + a_2b_1 - b_0a_3 \end{vmatrix}$$

and

$$L' = \begin{vmatrix} 0 & b_0 & b_2 \\ b_0 & b_1 & 0 \\ a_0b_1 & a_0b_2 + a_1b_1 & a_2b_2 - b_0a_4 \end{vmatrix}.$$

By again combining the two equations of the first system with both of the second system, we have the determinant

$$R = \begin{vmatrix} 0 & b_0 & b_1 & b_2 \\ b_0 & b_1 & b_2 & 0 \\ a_0b_1 & a_0b_2 + a_1b_1 & a_1b_2 + a_2b_1 - b_1a_3 & a_2b_2 - b_0a_4 \\ a_0b_2 & a_1b_2 & a_2b_2 - b_0a_4 & a_0b_2 - a_4b_1 \end{vmatrix}$$

which is the last simplified residue, or in other terms, the resultant to the system U, V .

Art. (7). It is most important to observe that the Bezoutian matrix to two functions of the same degree (n) is a symmetrical matrix, the terms similarly disposed in respect to one of the diagonals being equal.

Thus retaining the notation of Art. 5, so that

$$\begin{aligned}
(0, 1) &= a\beta - b\alpha, & (1, 2) &= b\gamma - c\beta, & (2, 3) &= c\delta - d\gamma, \\
(0, 2) &= a\gamma - c\alpha, & (1, 3) &= b\delta - d\beta, & \&c. \\
(0, 3) &= a\delta - d\alpha, & \&c.
\end{aligned}$$

when $n = 1$ the Bezoutian matrix consists of a single term $(0, 1)$; when $n = 2$, it becomes

$$\begin{pmatrix} (0, 1) & (0, 2) \\ (0, 2) & (1, 2) \end{pmatrix};$$

when $n = 3$, it becomes

$$\begin{array}{ccc} (0, 1) & (0, 2) & (0, 3) \\ (0, 2) & \left(\begin{array}{c} (0, 3) \\ + \\ (1, 2) \end{array} \right) & (1, 3) \\ (0, 3) & (1, 3) & (2, 3); \end{array}$$

p. 448

when $n = 4$, it becomes

$$\begin{array}{cccc} (0, 1) & (0, 2) & (0, 3) & (0, 4) \\ (0, 2) & \left(\begin{array}{c} (0, 3) \\ + \\ (1, 2) \end{array} \right) & \left(\begin{array}{c} (0, 4) \\ + \\ (1, 3) \end{array} \right) & (1, 4) \\ (0, 3) & \left(\begin{array}{c} (0, 4) \\ + \\ (1, 3) \end{array} \right) & \left(\begin{array}{c} (1, 4) \\ + \\ (2, 3) \end{array} \right) & (2, 4) \\ (0, 4) & (1, 4) & (2, 4) & (3, 4); \end{array}$$

when $n = 5$, it becomes

$$\begin{array}{ccccc} (0, 1) & (0, 2) & (0, 3) & (0, 4) & (0, 5) \\ (0, 2) & \left(\begin{array}{c} (0, 3) \\ + \\ (1, 2) \end{array} \right) & \left(\begin{array}{c} (0, 4) \\ + \\ (1, 3) \end{array} \right) & \left(\begin{array}{c} (0, 5) \\ + \\ (1, 4) \end{array} \right) & (1, 5) \\ (0, 3) & \left(\begin{array}{c} (0, 4) \\ + \\ (1, 3) \end{array} \right) & \left(\begin{array}{c} (0, 5) \\ + \\ (1, 4) \\ + \\ (2, 3) \end{array} \right) & \left(\begin{array}{c} (1, 5) \\ + \\ (2, 4) \end{array} \right) & (2, 5) \\ (0, 4) & \left(\begin{array}{c} (0, 5) \\ + \\ (1, 4) \end{array} \right) & \left(\begin{array}{c} (1, 5) \\ + \\ (2, 4) \end{array} \right) & \left(\begin{array}{c} (2, 5) \\ + \\ (3, 4) \end{array} \right) & (3, 5) \\ (0, 5) & (1, 5) & (2, 5) & (3, 5) & (4, 5), \end{array}$$

and so forth. Every such square it is apparent may be conceived as a sort of sloped pyramid, formed by the successive superposition of square layers, which layers possess not merely a simple symmetry about a diagonal (such as is proper to a *multiplication* table), but the higher symmetry (such as exists in an *addition* table), evinced in all the terms in any line of terms parallel to the diagonal transverse to the axis of symmetry being alike.² Thus for $n = 5$, the three layers

²A square arrangement having this kind of symmetry, namely, such as obtains in the so-called Pythagorean addition table as distinguished from that which obtains in the multiplication table, may be universally called Persymmetric.

or stages in question will be seen to be, the first—

$$\begin{array}{ccccc}
 (0, 1) & (0, 2) & (0, 3) & (0, 4) & (0, 5) \\
 (0, 2) & (0, 3) & (0, 4) & (0, 5) & (1, 5) \\
 (0, 3) & (0, 4) & (0, 5) & (1, 5) & (2, 5) \\
 (0, 4) & (0, 5) & (1, 5) & (2, 5) & (3, 5) \\
 (0, 5) & (1, 5) & (2, 5) & (3, 5) & (4, 5);
 \end{array}$$

the second—

$$\begin{array}{ccc}
 (1, 2) & (1, 3) & (1, 4) \\
 (1, 3) & (1, 4) & (2, 4) \\
 (1, 4) & (2, 4) & (3, 4);
 \end{array}$$

and the third—

$$(2, 3).$$

In general, when n is odd, say $2p + 1$, the pyramid will end with a single term $(p, (p + 1))$, and when even, as $2p$, with a square of four terms,

$$\begin{array}{ccc}
 ((p - 2), (p - 1)), & ((p - 2), p), \\
 ((p - 2), p), & ((p - 1), p).
 \end{array}$$

Each stage may be considered as consisting of three parts, a diagonal set of equal terms transverse to the axis of symmetry, and two triangular wings, one to the left, and the other to the right of this diagonal; the terms in each such diagonal for the respective stages will be

$$(0, n), \quad (1, n - 1), \quad (2, (n - 2)) \dots (p, (p + 1)),$$

p being $\frac{n}{2} - 1$ when n is even, and $\frac{n-1}{2}$ when n is odd.

If we change the order of the coefficients in each of the two given functions, it will be seen that the only effect will be to make the left and right triangular wings to change places, the diagonals in each stage remaining unaltered. The mode of forming these triangles is an operation of the most simple and mechanical nature, too obvious to need to be further insisted on here.

Art. (8). When we are dealing with two functions of unequal degrees, n and $n + e$, we can still form a square matrix with the coefficients of the two systems of e and n equations respectively, but this will no longer be symmetrical about a diagonal; it is obvious, however, that if we treat the function of the lower degree, as if it were of the same degree as the other function, which we may do by filling up the vacant places with terms affected with zero coefficients, the symmetry will be recovered; and it is somewhat important (as will appear hereafter) to compare the values of the Bezoutian secondaries as obtained, first in their simplest form by treating each of the two functions as complete in itself, and secondly, as they come out, when that of the functions which is of the lower degree is looked

upon as a defective form of a function of the same degree as the other. A single example will suffice to make the nature of the relation between the two sets of results apparent.

Take

$$fx = ax^4 + bx^3 + cx^2 + dx + e, \quad \phi x = 0x^4 + 0x^3 + \gamma x^2 + \delta x + \varepsilon.$$

p. 450

The general method of Art. 7 then gives for the Bezoutian matrix

$$\begin{array}{cccc} 0 & a\gamma & a\delta & a\varepsilon \\ a\gamma & \begin{pmatrix} a\delta \\ + \\ b\gamma \end{pmatrix} & \begin{pmatrix} a\varepsilon \\ + \\ b\delta \end{pmatrix} & b\varepsilon \\ a\delta & \begin{pmatrix} a\varepsilon \\ + \\ b\delta \end{pmatrix} & \begin{pmatrix} b\varepsilon \\ + \\ c\delta - d\gamma \end{pmatrix} & c\varepsilon - e\gamma \\ a\varepsilon & b\varepsilon & c\varepsilon - e\gamma & d\varepsilon - e\delta. \end{array}$$

We shall not affect the value either of the complete determinant, or of any of the minor determinants appertaining to the above matrix, by subtracting the first line of terms, each increased in the ratio of $b : a$, from the second line of terms respectively; the matrix so modified becomes

$$\begin{array}{cccc} 0 & a\gamma & a\delta & a\varepsilon \\ a\gamma & a\delta & a\varepsilon & 0 \\ a\delta & a\varepsilon + b\delta & \begin{pmatrix} b\varepsilon \\ + \\ c\delta - d\gamma \end{pmatrix} & c\varepsilon - e\gamma \\ a\varepsilon & b\varepsilon & c\varepsilon - e\gamma & d\varepsilon - e\delta. \end{array}$$

Again, adopting the method of Art. 6, we should obtain the matrix

$$\begin{array}{cccc} 0 & \gamma & \delta & \varepsilon \\ \gamma & \delta & \varepsilon & 0 \\ \delta & a\varepsilon - b\delta & \begin{pmatrix} b\varepsilon \\ + \\ c\delta - d\gamma \end{pmatrix} & c\varepsilon - e\gamma \\ a\varepsilon & b\varepsilon & c\varepsilon - e\gamma & d\varepsilon - e\delta. \end{array}$$

Hence it is apparent that the secondary Bezoutians obtained by the symmetrizing method will differ from those obtained by the unsymmetrical method by a constant factor a^2 ; and so in general it may readily be shown that the secondary Bezoutians, by the use of the symmetrizing method, will each become affected with a constant irrelevant factor a^ω , where ω is the difference of the degrees of the two functions, and a the leading coefficient of the higher one of the two.

When a is taken unity, the Bezoutian secondaries, as obtained by either method, will of course be identical.

Art. (9). There is another method³ of obtaining the simplified residues to any two functions U and V of the degrees n and $n + e$ respectively, which, p. 451

although less elegant, ought not to be passed over in silence. This method consists in forming the identical equations (of which for greater brevity the right-hand members are suppressed)

$$\begin{aligned}
 V &= \&c. \\
 xV &= \&c. \\
 &\vdots \\
 x^{e-1}V &= \&c. \\
 U &= \&c. \\
 x^eV &= \&c. \\
 xU &= \&c. \\
 x^{e+1}V &= \&c. \\
 x^2U &= \&c. \\
 x^{e+2}V &= \&c. \\
 \&c. &= \&c. \\
 x^{n-1}U &= \&c. \\
 x^{e+n-1}V &= \&c.
 \end{aligned}$$

If we equate the right-hand members of $(e + 2\iota)$ of the above equations to zero, and then eliminate dialytically the several powers of x from $x^{n+e+\iota-1}$ to $x^{n+\iota+1}$ (both inclusive), the result of this process will evidently be of $(e + \iota)$ dimensions in respect of the coefficients in V , of ι dimensions in respect of the coefficients in U and of the degree $x^{n-\iota}$ in x ; it will also be of the form

$$(A + Bx + \cdots + Lx^{\iota-1})U + (F + Gx + \cdots + Qx^{e+\iota-1})V,$$

and by virtue of Art. 2, must consequently be the ι th simplified residue to the system U, V .

Art. (10). The most general view of the subject of expansion by the method of continued division, consists in treating the process as having reference solely to the two systems of coefficients in U and V , which themselves are to be regarded in the light of generating functions. To carry out this conception, we ought to write

$$\begin{aligned}
 U &= a_0 + a_1y + a_2y^2 + a_3y^3 + \&c. \text{ ad inf.}, \\
 V &= b_0 + b_1y + b_2y^2 + b_3y^3 + \&c. \text{ ad inf.},
 \end{aligned}$$

³Originally given by myself in the *London and Edinburgh Philosophical Magazine*, as long ago as 1839 or 1840 [p. 54 above]; and some years subsequently in unconsciousness of that fact, reproduced by my friend Mr Cayley, to whom the method is sometimes erroneously ascribed, and who arrived at the same equations by an entirely different circle of reasoning.

and might then suppose the process of successive division applied to U and V , so as to obtain the successive equations

$$\begin{aligned} U - M_1V + R_1 &= 0, \\ V - M_2R_1 + R_2 &= 0, \\ R_1 - M_3R_2 + R_3 &= 0, \\ &\&c. \quad \&c., \end{aligned}$$

$M_1, M_2, M_3, \&c.$ being each severally of any degree whatever in y , and in general the degree of y in M_ι being any given arbitrary function $\phi(\iota)$ of ι . The values of the coefficients of the residues $R_1, R_2, R_3 \dots$, or of these forms simplified by the rejection of detachable factors, become then the distinct object of the inquiry, and will, of course, depend only upon the coefficients in U and V and the nature of the arbitrary continuous or discontinuous function $\phi(\iota)$, which regulates the number of *steps* through which each successive process of division is to be pursued. Following out this idea in a particular case, if we again reduce our two initial functions to the forms previously employed, and write

$$U = a_0x^n + a_1x^{n-1} + \&c., \quad V = b_0x^n + b_1x^{n-1} + \&c.;$$

and if, instead of making, according to the more usual course of proceeding, the divisions proceed first through one step and ever after through two steps at a time, which is tantamount to making $\phi 1 = 1, \phi(1 + \omega) = 2$, we push each division through one step only at a time, and no more (so that in fact $\phi(\iota)$ is always 1), we shall have

$$\begin{aligned} U - m_1V + R_1 &= 0, \\ V - m_2xR_1 + R_2 &= 0, \\ R_1 - m_3R_2 + R_3 &= 0, \\ R_2 - m_4xR_3 + R_4 &= 0, \\ &\&c. \quad \&c., \end{aligned}$$

$m_1, m_2, m_3, \&c.$ being functions of the coefficients only of U and V ; and it is not without interest to observe (which is capable of an easy demonstration) that the simplified residues contained in $R_1, R_2, \&c.$, found according to this mode of development, will be the successive dialytic resultants obtained by eliminating the $(\iota - 1)$ th highest powers of x between the ι first of the system of annexed

equations (supposed to be expressed in terms of x)

$$\begin{aligned}
 U &= 0, \\
 V &= 0, \\
 xU &= 0, \\
 xV &= 0, \\
 x^2U &= 0, \\
 x^2V &= 0, \\
 \&c. &= \&c., \\
 x^{n-1}U &= 0, \\
 x^{n-1}V &= 0.
 \end{aligned}$$

p. 453

If we combine together $2i + 1$ of the above equations, the highest power of x entering on the left-hand side will be x^{n+i} , and we shall be able to eliminate $2i$ of these factors, leaving x^{n-i} the highest power remaining uneliminated. If we take $2i$, that is i pairs of the equations, the highest power of x appearing in any of them will be x^{n+i-1} , and we shall be able to eliminate between them so as still to leave $x^{n+i-1-(2i-1)}$, that is x^{n-i} as before, the highest power of x remaining uneliminated; and it will be readily seen that such of the simplified residues corresponding to this mode of development as occupy the odd places in the series of such residues, will be identical with the successive simplified residues resulting from the ordinary mode of developing $\frac{U}{V}$ under the form of a continued fraction.

Art. (11). It has been shown that the simplified residues of fx and ϕx resulting from the process of continued division are identical in point of form with the secondary Bezoutians of these functions, but it remains to assign the numerical relations between any such residue and the corresponding secondary.

To determine this numerical relation, it will of course be sufficient to compare the magnitude of the coefficient of any one power of x in the one, with that of the same power in the other; and for this purpose I shall make choice of the leading coefficients in each. In what follows, and throughout this paper, it will always be understood that in calculating the determinant corresponding to any square the product of the terms situated in the diagonal descending from left to right will always be taken with the positive sign, which convention will serve to determine the sign of all the other products entering into such determinant. Now adopting the umbral notation for determinants,⁴ we have, by virtue of a much more general theorem for compound determinants, the following identical equation:

$$\begin{aligned}
 &\left(\begin{smallmatrix} a_1 a_2 a_3 \dots a_{m-1} \\ \alpha_1 \alpha_2 \alpha_3 \dots \alpha_{m-1} \end{smallmatrix} \right) \times \left(\begin{smallmatrix} a_1 a_2 a_3 \dots a_{m+1} \\ \alpha_1 \alpha_2 \alpha_3 \dots \alpha_{m+1} \end{smallmatrix} \right) \\
 &= \left(\begin{smallmatrix} a_1 a_2 a_3 \dots a_{m-1} a_m \\ \alpha_1 \alpha_2 \alpha_3 \dots \alpha_{m-1} \alpha_m \end{smallmatrix} \right) \times \left(\begin{smallmatrix} a_1 a_2 \dots a_{m-1} a_{m+1} \\ \alpha_1 \alpha_2 \dots \alpha_{m-1} \alpha_{m+1} \end{smallmatrix} \right) - \left(\begin{smallmatrix} a_1 a_2 a_3 \dots a_{m-1} a_m \\ \alpha_1 \alpha_2 \dots \alpha_{m-1} \alpha_{m+1} \end{smallmatrix} \right) \times \left(\begin{smallmatrix} a_1 a_2 \dots a_{m-1} a_{m+1} \\ \alpha_1 \alpha_2 \dots \alpha_{m-1} \alpha_m \end{smallmatrix} \right),
 \end{aligned}$$

and consequently

$$\left(\begin{smallmatrix} a_1 a_2 a_3 \dots a_{m-1} \\ \alpha_1 \alpha_2 \alpha_3 \dots \alpha_{m-1} \end{smallmatrix} \right) \times \left(\begin{smallmatrix} a_1 a_2 a_3 \dots a_{m-1} a_m a_{m+1} \\ \alpha_1 \alpha_2 \alpha_3 \dots \alpha_{m-1} a_m a_{m+1} \end{smallmatrix} \right)$$

⁴See *London and Edinburgh Philosophical Magazine*, April 1851 [p. 242 above].

$$= \begin{pmatrix} a_1 a_2 a_3 \dots a_{m-1} a_m \\ \alpha_1 \alpha_2 \alpha_3 \dots \alpha_{m-1} \alpha_m \end{pmatrix} \times \begin{pmatrix} a_1 a_2 \dots a_{m-1} a_{m+1} \\ \alpha_1 \alpha_2 \dots \alpha_{m-1} \alpha_{m+1} \end{pmatrix} - \begin{pmatrix} a_1 a_2 \dots a_{m-1} a_m \\ \alpha_1 \alpha_2 \dots \alpha_{m-1} \alpha_{m+1} \end{pmatrix}^2,$$

p. 454

and consequently when

$$\begin{pmatrix} a_1 a_2 \dots a_{m-1} a_m \\ \alpha_1 \alpha_2 \dots \alpha_{m-1} \alpha_{m+1} \end{pmatrix} = 0,$$

$$\begin{pmatrix} a_1 a_2 \dots a_{m-1} \\ \alpha_1 \alpha_2 \dots \alpha_{m-1} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} a_1 a_2 \dots a_{m-1} a_m a_{m+1} \\ \alpha_1 \alpha_2 \dots \alpha_{m-1} \alpha_m \alpha_{m+1} \end{pmatrix}$$

will have different algebraical signs, it being of course understood that all the quantities entering into the determinants thus umbrally represented above are supposed to be real quantities. This theorem, translated into the ordinary language of determinants, may be stated as follows:—Begin with any square of terms whether symmetrical or otherwise, say of r lines and r columns: let this square be bordered laterally and longitudinally by the same number r of new quantities symmetrically disposed in respect to one of the diagonals, the term common to the superadded line and column being filled up with any quantity whatever; we thus obtain a square of $(r + 1)$ lines and columns; let this be again bordered laterally and longitudinally by $(r + 1)$ quantities symmetrically disposed above the same diagonal as that last selected, the place in which this new line and column meet being also filled up with any arbitrary quantity; and proceeding in this manner, let the determinants corresponding to the square matrices thus formed be called $D_r, D_{r+1}, D_{r+2} \dots$: this series of quantities will possess the property, that no term in it can vanish without the terms on either side of that so vanishing having contrary signs. Thus if we begin with a square consisting of one single term, we may suppose that by accretions formed after the above rule it has been developed into the square (M) below written, and which of course may be indefinitely extended:

$$\begin{array}{ccccc} a & l & m & p & s \\ l & b & n & q & t \\ m & n & c & r & u \\ p & q & r & d & v \\ s & t & u & v & e \end{array} \quad (\text{M})$$

Here $D_0, D_1, D_2, D_3, D_4, D_5$ will represent the progression

$$1, \quad a, \quad \begin{vmatrix} a & l \\ l & b \end{vmatrix}, \quad \begin{vmatrix} a & l & m \\ l & b & n \\ m & n & c \end{vmatrix}, \quad \begin{vmatrix} a & l & m & p \\ l & b & n & q \\ m & n & c & r \\ p & q & r & d \end{vmatrix}, \quad \begin{vmatrix} a & l & m & p & s \\ l & b & n & q & t \\ m & n & c & r & u \\ p & q & r & d & v \\ s & t & u & v & e \end{vmatrix};$$

(II) p. 455

so if we use the matrix

$$\begin{array}{ccccc} a & l & m & p & s \\ l' & b & n & q & t \\ m & n & c & r & u \\ p & q & r & d & v \\ s & t & u & v & e \end{array}$$

the determinants D_1, D_2, D_3, D_4 , representing

$$a, \quad \begin{vmatrix} a & l \\ l' & b \end{vmatrix}, \quad \begin{vmatrix} a & l & m \\ l' & b & n \\ m & n & c \end{vmatrix}, \quad \begin{vmatrix} a & l & m & p \\ l' & b & n & q \\ m & n & c & r \\ p & q & r & d \end{vmatrix},$$

will possess the property in question; the line and column l, b ; l', b not being identical, the first determinant D_0 representing unity must not be included in the progression.

We shall have occasion to use this theorem as applicable to the case of a matrix symmetrical throughout, and we may term the progression (II), above written, a progression of the successive principal determinants about the axis of symmetry of the square matrix (M), and so in general. Now it is obvious that the leading coefficients of the successive Bezoutian secondaries are the successive principal determinants about the axis of symmetry of the Bezoutian squares; they will therefore have the property which has been demonstrated of such progressions; to wit, if the first of them vanishes, the second will have a sign contrary to that of $+1$; if the second vanishes, the third will have a sign contrary to that of the first, and so on.

Art. (12). Now let fx and ϕx be any two algebraical functions of x with the leading coefficients in each, for greater simplicity, supposed positive: and in the course of developing $\frac{\phi x}{fx}$ under the form of an improper continued fraction by the common process of successive division, let any two consecutive residues (the word residue being used in the same conventional sense as employed throughout) be

$$\begin{aligned} Ax^t + Bx^{t-1} + Cx^{t-2} + \&c. \\ B'x^{t-1} + C'x^{t-2} + D'x^{t-3} + \&c. \end{aligned}$$

The residue next following, obtained by actually performing the division and duly changing the sign of the remainder, will be

$$\left\{ \left(\frac{AD'}{B'} - C \right) - \left(\frac{AC'}{B'} - B \right) \frac{C'}{B'} \right\} x^{t-2} + \&c.,$$

which is of the form

$$\frac{1}{B'^2} \{ B'M - AC'^2 \} x^{t-2} + \&c.$$

Thus the leading coefficients in the complete unreduced residues will be

$$A, \quad B', \quad \frac{1}{B'^2}\{B'M - AC'^2\},$$

and when reduced by the expulsion of the allotropic factor will become

$$A, \quad B', \quad B'M - AC'^2,$$

and consequently, when B' the leading coefficient of one of the simplified residues vanishes, the leading coefficients of the residues immediately preceding and following that one will have contrary signs.

First, let fx and ϕx be of the same degree. As regards the numerical ratio of each Bezoutian secondary to the corresponding simplified residue, it has been already observed that there are always unit coefficients in the latter of these, and the same is obviously true of the former; hence if we call the progression of the leading coefficients of the simplified residues

$$R_1, R_2, R_3, R_4, \&c.,$$

and that of the leading coefficients of the Bezoutian secondaries

$$B_1, B_2, B_3, B_4, \&c.,$$

we have

$$B_1 = \pm R_1, \quad B_2 = \pm R_2, \quad B_3 = \pm R_3, \quad B_4 = \pm R_4, \quad \&c.$$

It may be proved by actual trial that $B_1 = R_1$ and $B_2 = R_2$. Moreover, since the signs are invariable, and do not depend upon the values of the coefficients, we may suppose $B_2 = 0$ (which may always be satisfied by real values of the quantities of which B_2 is a function); we shall also, therefore, have $R_2 = 0$, and consequently B_3 has the opposite sign to that of B_1 , and R_3 the opposite sign to that of R_1 , which is equal to B_1 : hence when $B_2 = 0$, B_3 and R_3 are equal, and consequently are always equal; in like manner we can prove that R_4 and B_4 have the same sign when R_3 and B_3 vanish, and consequently are always equal, and so on *ad libitum*, which proves that the series $B_1, B_2, \dots B_n$ is identical with the series $R_1, R_2, \dots R_n$, and consequently that the Bezoutian secondaries are identical in form, magnitude and algebraical sign with the simplified residues.

Secondly, when fx and ϕx are not of the same degree, it has been shown that the secondaries formed from the non-symmetrical matrix corresponding to this case will be the same as those formed from the *symmetrical* matrix corresponding to fx and Φx (where Φx is ϕx treated by aid of evanescent terms as of the same degree as fx), with the exception merely of a constant multiplier (a power of the leading coefficient of fx) being introduced into each secondary. By aid of this observation, the proposition

established for the case of two functions of the same degree may be readily seen to be capable of being extended, from the case of f and ϕ being of equal dimensions in x , to the general case of their dimensions being any whatever.

Art. (13). Before closing this section, it may be well to call attention to the nature of the relation which connects the successive residues of fx and ϕx with these functions themselves, and with the improper continued fractional form into which $\frac{\phi x}{fx}$ is supposed to be developed in the process of obtaining these residues.

If ϕx be of n degrees, and fx of $n + e$ degrees in x , we shall have

$$\frac{\phi x}{fx} = \frac{1}{Q_1 - \frac{1}{q_2 - \frac{1}{q_3 - \cdots - \frac{1}{q_n}}}},$$

where Q_1 may be supposed to be a function of x of the degree e , and $q_2, q_3, \dots, q_n$, are all linear functions of x ; the total number of the quotients $Q_1, q_2, \dots, q_n$ being of course n when the process of continued division is supposed to be carried out until the last residue is zero. Upon this supposition the last but one residue is a constant, the preceding one a function of x of the first degree, the one preceding that a function of x of the second degree, and so on.

Let us call the residue of the degree ι in x , $\mathfrak{R}_\iota$; it will readily be seen that the successive complete residues arranged in an ascending order will be

$$\mathfrak{R}_0, \quad \mathfrak{R}_0 q_n, \quad \mathfrak{R}_0(q_{n-1}q_n - 1), \quad \mathfrak{R}_0(q_{n-2}q_{n-1}q_n - q_{n-2} - q_n), \quad \&c.,$$

being in the ratios

$$1, \quad q_n, \quad q_{n-1} - \frac{1}{q_n}, \quad q_{n-2} - \frac{1}{q_{n-1} - \frac{1}{q_n}}, \quad \&c.$$

Again, we shall have in general

$$\Lambda_\iota f - L_\iota \phi = \mathfrak{R}_\iota, \tag{15}$$

Λ_ι being an integral function of x of the degree $n - \iota - 1$, and L_ι an integral function of x of the degree $(n + e) - \iota - 1$; and it is easy to see that the successive convergents to the continued fraction

$$\frac{1}{Q_1 - \frac{1}{q_2 - \frac{1}{q_3 - \&c.}}}$$

have their respective numerators and denominators identical with those of the fractions

$$\frac{\Lambda_{n-1}}{L_{n-1}}, \quad \frac{\Lambda_{n-2}}{L_{n-2}}, \quad \frac{\Lambda_{n-3}}{L_{n-3}}, \quad \&c.$$